

10.3 Matrix Representations of Graphs

An $m \times n$ (read “ m by n ”) **matrix $\mathbf{A}$ over a set S** is a rectangular array of elements of S arranged into m rows and n columns.

We write $\mathbf{A} = (a_{ij})$

10.3.1 Adjacency Matrix

Let G be a directed graph with ordered vertices $v_1, v_2, \dots, v_n$. The adjacency matrix of G is the $n \times n$ matrix $\mathbf{A} = (a_{ij})$ over the set of nonnegative integers such that

$$a_{ij} = \text{the number of arrows from } v_i \text{ to } v_j \text{ for all } i, j = 1, 2, \dots, n. \quad (1)$$

10.3.2 Adjacency Matrix

Let G be an undirected graph with ordered vertices $v_1, v_2, \dots, v_n$. The adjacency matrix of G is the $n \times n$ matrix $A = (a_{ij})$ over the set of nonnegative integers such that

$$\begin{aligned} a_{ij} &= \text{the number of edges connecting } v_i \text{ and } v_j \\ &\text{for all } i, j = 1, 2, \dots, n. \end{aligned} \quad (2)$$

10.3.3 Symmetric

An $n \times n$ square matrix $A = (a_{ij})$ is called **symmetric** if, and only if, for all $i, j = 1, 2, \dots, n$,

$$a_{ij} = a_{ji} \quad (3)$$

10.3.4 Identity Matrix

For each positive integer n , the $n \times n$ **identity matrix**, denoted $I_n = (\delta_{ij})$ or just **I** (if the size of the matrix is obvious from context), is the $n \times n$ matrix in which all the entries in the main diagonal are 1's and all other entries are 0's. In other words,

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \text{for all } i, j = 1, 2, \dots, n$$

10.3.5 The Number of Walks

If G is a graph with vertices $v_1, v_2, \dots, v_m$ and A is the adjacency matrix of G , then for each positive integer n and for all integers $i, j = 1, 2, \dots, m$,

the ij th entry of A_n = the number of walks of length n from v_i to v_j .

10.5 Trees

10.5.1 Terminology

A graph is said to be **circuit-free** if, and only if, it has no circuits. A graph is called a **tree** if, and only if, it is circuit-free and connected. A **trivial tree** is a graph that consists of a single vertex. A graph is called a **forest** if, and only if, it is circuit-free and not connected.

10.5.2 Vertex-Degree Relation

Any tree that has more than one vertex has at least one vertex of degree 1.

10.5.3 Terminal Vertex

Let T be a tree. If T has only one or two vertices, then each is called a **terminal vertex**. If T has at least three vertices, then a vertex of degree 1 in T is called a **terminal vertex** (or a **leaf**), and a vertex of degree greater than 1 in T is called an **internal vertex** (or a **branch vertex**).

10.5.4 Edges And Vertices

For any positive integer n , any tree with n vertices has $n - 1$ edges.

10.6 Connected, Circuit

If G is any connected graph, C is any circuit in G , and any one of the edges of C is removed from G , then the graph that remains is connected.